

Executive Summary: The End-to-End Sales Forecasting & Demand Intelligence System was developed to transition our supply chain strategy from reactive to predictive. By analyzing four years of daily historical data, we successfully engineered an XGBoost machine learning pipeline that forecasts future category demand with a highly accurate 13.07% error rate. Furthermore, we implemented automated anomaly detection to flag unexpected supply shocks and segmented our product catalog into four distinct demand profiles to optimize warehouse capital.

Key Findings & Forecasts

- **EDA Highlights:** The West region demonstrates the highest and most consistent sales growth. Additionally, November and December exhibit massive, repeating seasonal spikes tied to holiday shopping.
- **3-Month Forecast:** Our champion XGBoost model projects a highly volatile upcoming quarter, with an initial drop in Month 2 followed by a massive surge to \$59,629 in Month 3. The Technology category and West Region are projected to drive the majority of this growth.

Anomaly Detection & Segmentation Strategy

- **Anomalies:** The system flagged major global spikes specifically during late November and early December. These correspond to Black Friday and Q4 corporate budget exhaustion. Localized mid-year spikes (e.g., March) were identified as likely bulk B2B enterprise orders.
- **Stocking Strategy:** Through K-Means clustering, we identified four product segments. High-margin, growing items like *Copiers* require premium safety stock to prevent expensive stock-outs. Conversely, stable baseline items like *Paper* should be moved to fully automated algorithmic reordering, while flat items like *Fasteners* should be aggressively minimized in the warehouse.

Business Recommendations & System Limitations

1. **Capital Reallocation:** Shift warehouse floor space allocation heavily toward the West Region and Technology category prior to Month 3 to capture the forecasted growth surge.
 2. **Dynamic Safety Stock:** Implement a Just-In-Time (JIT) agile ordering system for high-volume/high-volatility items (e.g., Phones, Chairs) to prevent overstocking during non-Q4 months.
 3. **Third-Party Logistics:** Transition declining, low-volume clusters (e.g., Fasteners, Labels) to a direct-to-consumer drop-shipping model to eliminate dead holding costs.
- **System Risk:** The core limitation of this machine learning system is its reliance on historical stationarity. The model cannot account for sudden, unprecedented macroeconomic shocks (e.g., global shipping lane closures or sudden inflation spikes) that exist outside the trained data matrix.